Math 110 — Week 10

Orthonormal bases, Gram–Schmidt, orthogonal complements and projections (Notes Ch. 18, LADR 6B–6C, HW10)

§1 Facts used from earlier weeks

V is an inner product space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. When V is finite-dimensional, $n = \dim V$.

(T1) **Rules of the inner product** (Week 9). $\langle \lambda u + v, w\rangle = \lambda\langle u, w\rangle + \langle v, w\rangle$, $\langle u, \lambda v + w\rangle = \bar{\lambda}\langle u, v\rangle + \langle u, w\rangle$, $\langle u, v\rangle = \overline{\langle v, u\rangle}$, $\langle v, v\rangle > 0$ for $v \neq 0$, and $\|v\| = \sqrt{\langle v, v\rangle}$.

(T2) **Pythagoras** (Week 9). If $\langle u, v\rangle = 0$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

(T3) **Annihilators** (Week 5). For a subspace U of V,

$$U^0 = \{\varphi \in V' : \varphi(u) = 0 \text{ for all } u \in U\},$$

and $\dim U^0 = n - \dim U$; also $\dim V' = \dim V$.

(T4) **Direct sums** (Week 1). $V = U \oplus W$ means $U + W = V$ and $U \cap W = \{0\}$; then every v is $u + w$ ($u \in U$, $w \in W$) in exactly one way, and $\dim(U \oplus W) = \dim U + \dim W$ (Week 2: $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$).

(T5) **Dimension shortcuts** (Week 2). An independent list of n vectors is a basis; a subspace of dimension n is V; an injective linear map between spaces of the same finite dimension is an isomorphism.

§2 This week

§2.1 Orthonormal lists (Ch. 18)

The question. In $\mathbb{R}^n$ the coordinates of a vector in the standard basis can be read off without solving any equations. In which bases of an inner product space is that possible, and what does a norm look like in such a basis?

Definition (Def. 46 / LADR 6.22)**.** u, v are **orthogonal** if $\langle u, v\rangle = 0$. A list $e_1, \dots, e_m$ is **orthonormal** if $\langle e_i, e_j\rangle = \delta_{ij}$: each e_j has norm 1 and any two different ones are orthogonal.

Examples of orthonormal lists. The standard basis of $\mathbb{F}^n$; $(1, 1)/\sqrt{2}$, $(1, -1)/\sqrt{2}$ in $\mathbb{R}^2$; and in the L^2 product on $[0, 1]$, the functions $e^{2\pi i n x}$ ($n \in \mathbb{Z}$), in the sense that every finite sublist is orthonormal, because $\int_0^1 e^{2\pi i(n-k)x}\,dx$ is 1 for $n = k$ and 0 otherwise.

Theorem (Prop. 84 / LADR 6.24: the norm in an orthonormal list is a sum of squares)
If $e_1, \dots, e_m$ is orthonormal, then $\|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2$ for all scalars a_j.

The idea

This is Pythagoras applied $m-1$ times. The last term $a_m e_m$ is orthogonal to the sum of the others, so the squared norm splits into the squared norm of the first $m-1$ terms plus $|a_m|^2$; repeat.

Proof. Induction on m.

1. For $m = 1$: $\|a_1 e_1\|^2 = |a_1|^2 \|e_1\|^2 = |a_1|^2$.
2. Let $m \geq 2$ and $w = a_1 e_1 + \cdots + a_{m-1} e_{m-1}$. Then $\langle w, e_m \rangle = \sum_{j<m} a_j \langle e_j, e_m \rangle = 0$, so $\langle w, a_m e_m \rangle = \bar{a}_m \langle w, e_m \rangle = 0$: w and $a_m e_m$ are orthogonal.
3. By Pythagoras and the induction hypothesis,

$$\|w + a_m e_m\|^2 = \|w\|^2 + \|a_m e_m\|^2 = |a_1|^2 + \cdots + |a_{m-1}|^2 + |a_m|^2.$$

□

Theorem (Prop. 85 / LADR 6.25: orthonormal lists are independent)
Every orthonormal list is linearly independent.

Proof. If $a_1 e_1 + \cdots + a_m e_m = 0$, Prop. 84 gives $0 = \|0\|^2 = |a_1|^2 + \cdots + |a_m|^2$. A sum of nonnegative numbers is 0 only if each is 0, so every $a_j = 0$. □

In particular, an orthonormal list of $\dim V$ vectors is an orthonormal basis: it is independent (Prop. 85) and has the right length (T5).

Theorem (Prop. 86 / LADR 6.30: coordinates in an orthonormal basis are inner products)
If $e_1, \dots, e_n$ is an orthonormal basis, then for every v,

$$v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_n \rangle e_n, \qquad \|v\|^2 = |\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_n \rangle|^2,$$

and for all u, v,

$$\langle u, v \rangle = \langle u, e_1 \rangle \overline{\langle v, e_1 \rangle} + \cdots + \langle u, e_n \rangle \overline{\langle v, e_n \rangle}.$$

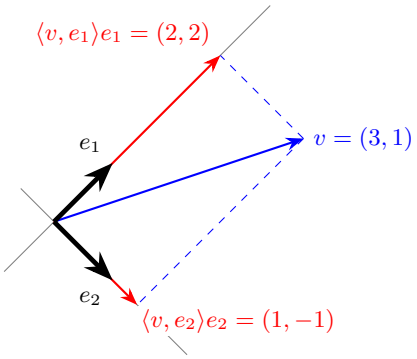
The idea

To find the coefficient a_k in $v = \sum a_j e_j$, take the inner product with e_k. Every term with $j \neq k$ gives 0, and the term $j = k$ gives a_k. So the coefficients are found one at a time, with no linear system to solve.

Proof. 1. $e_1, \dots, e_n$ is a basis, so $v = a_1 e_1 + \cdots + a_n e_n$ for some scalars.

2. Take the inner product with e_k and use linearity in the first slot: $\langle v, e_k \rangle = \sum_j a_j \langle e_j, e_k \rangle = a_k$, because $\langle e_j, e_k \rangle = 0$ for $j \neq k$ and 1 for $j = k$.
3. Substituting $a_k = \langle v, e_k \rangle$ gives the formula for v, and Prop. 84 gives the formula for $\|v\|^2$.
4. For $\langle u, v \rangle$, write $u = \sum_j \langle u, e_j \rangle e_j$ and $v = \sum_k \langle v, e_k \rangle e_k$. Expanding by linearity in the first slot and conjugate-linearity in the second, the term for (j, k) is $\langle u, e_j \rangle \overline{\langle v, e_k \rangle} \langle e_j, e_k \rangle$, and only the terms with $j = k$ are nonzero.

□



$v = (3,1)$ in the orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1,1)$, $e_2 = \frac{1}{\sqrt{2}}(1,-1)$. The coefficients are $\langle v, e_1\rangle = 2\sqrt{2}$ and $\langle v, e_2\rangle = \sqrt{2}$: the feet of the perpendiculars from v to the two lines. And $\|v\|^2 = 10 = (2\sqrt{2})^2 + (\sqrt{2})^2$.

Theorem (LADR 6.26: Bessel's inequality)
If $e_1, \dots, e_m$ is an orthonormal list (not necessarily a basis), then $|\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2 \le \|v\|^2$ for every v.

Proof. 1. Let $u = \sum_j \langle v, e_j\rangle e_j$ and $w = v - u$. For each k,

$$\langle w, e_k\rangle = \langle v, e_k\rangle - \sum_j \langle v, e_j\rangle\langle e_j, e_k\rangle = 0.$$

So w is orthogonal to u: for $u = \sum c_k e_k$, $\langle w, u\rangle = \sum \bar{c}_k \langle w, e_k\rangle = 0$.

2. By Pythagoras and Prop. 84,

$$\|v\|^2 = \|u\|^2 + \|w\|^2 \ge \|u\|^2 = \sum_j |\langle v, e_j\rangle|^2.$$

□

Example: Parseval for $f(x) = x$ (cited). Parseval's theorem (an infinite-dimensional version, for comparison). In the L^2 product on $[0,1]$ the functions $e_n(x) = e^{2\pi i n x}$, $n \in \mathbb{Z}$, are orthonormal. For a function f, the numbers $c_n = \langle f, e_n\rangle$ are its **Fourier coefficients**, and Parseval's theorem says $\sum_n |c_n|^2 = \|f\|^2$, the analogue of Prop. 86 with infinitely many terms. (Proving it needs convergence arguments that are not part of this course.) For $f(x) = x$:

$$c_0 = \int_0^1 x\,dx = \frac{1}{2}, \qquad c_n = \int_0^1 x e^{-2\pi i n x}\,dx = \frac{i}{2\pi n} \quad (n \ne 0), \qquad \|f\|^2 = \int_0^1 x^2\,dx = \frac{1}{3}.$$

So $|c_n|^2 = 1/(4\pi^2 n^2)$ for $n \ne 0$, and each $n \ge 1$ appears twice (as n and $-n$). Parseval gives

$$\frac{1}{4} + 2\sum_{n=1}^{\infty} \frac{1}{4\pi^2 n^2} = \frac{1}{3}, \qquad \text{so} \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = 2\pi^2\Big(\frac{1}{3} - \frac{1}{4}\Big) = \frac{\pi^2}{6}.$$

The question answered. In an orthonormal basis: the coordinates are the inner products $\langle v, e_k\rangle$, and $\|v\|^2$ is the sum of their squared absolute values (Prop. 86).

Shortcuts

- **Coordinates by inner products**: in an orthonormal basis, $a_k = \langle v, e_k\rangle$ and $\|v\|^2 = \sum |a_k|^2$. Reason: Prop. 86. Condition: orthonormal basis. Instance: $v = (3,1)$, $e_{1,2} = \frac{1}{\sqrt{2}}(1, \pm 1)$: $a = (2\sqrt{2}, \sqrt{2})$, $\|v\|^2 = 8 + 2$.
- **Orthogonal (not normalized) basis** f_k: $v = \sum_k \frac{\langle v, f_k\rangle}{\|f_k\|^2} f_k$. Reason: Prop. 86 for $e_k = f_k/\|f_k\|$. Instance: $f_1 = (1,1)$, $f_2 = (1,-1)$, $v = (3,1)$: $v = \frac{4}{2} f_1 + \frac{2}{2} f_2 = 2f_1 + f_2$.

- **Is it a basis?** An orthonormal list of $\dim V$ vectors is a basis. Reason: Prop. 85 and (T5). Instance: $(1,1)/\sqrt{2}$, $(1,-1)/\sqrt{2}$ in $\mathbb{R}^2$.
- **Matrix entries**: in an orthonormal basis, $\mathcal{M}(T)_{jk} = \langle Te_k, e_j\rangle$. Reason: column k holds the coordinates of Te_k, which are $\langle Te_k, e_j\rangle$ by Prop. 86. Instance: for the rotation by $\pi/2$ in the standard basis, $\mathcal{M}_{12} = \langle Te_2, e_1\rangle = \langle(-1,0), e_1\rangle = -1$.

Traps

- $a_k = \langle v, e_k\rangle$ needs an orthonormal basis: for the basis $(1,0), (1,1)$ and $v = (0,1)$, $\langle v, (1,0)\rangle = 0$, but $v = -(1,0) + (1,1)$.
- Orthogonal is not orthonormal: with $f_1 = (1,1)$, $\langle(3,1), f_1\rangle = 4$, but the coefficient of f_1 is 2.

§2.2 The Gram–Schmidt procedure (Ch. 18)

The question. Orthonormal bases make coordinates and norms easy. Does every finite-dimensional inner product space have one, and how do we find it from an arbitrary basis?

Theorem (Prop. 87 / LADR 6.32: Gram–Schmidt)
Let $v_1, \dots, v_n$ be a basis of V. Define $e_1 = v_1/\|v_1\|$ and, for $j = 2, \dots, n$,

$$e_j = \frac{w_j}{\|w_j\|}, \qquad w_j = v_j - \langle v_j, e_1\rangle e_1 - \cdots - \langle v_j, e_{j-1}\rangle e_{j-1}.$$

Then $e_1, \dots, e_n$ is an orthonormal basis of V, and $\operatorname{span}(e_1, \dots, e_j) = \operatorname{span}(v_1, \dots, v_j)$ for every j.

The idea

Build the basis one vector at a time. Given orthonormal $e_1, \dots, e_{j-1}$, the next vector v_j is generally not orthogonal to them. Remove from v_j its component along each e_k, which is $\langle v_j, e_k\rangle e_k$ by the idea of Prop. 86; what is left, w_j, is orthogonal to all of them. Then divide by the length. The only thing that could go wrong is $w_j = 0$, and that would mean v_j was a combination of the earlier vectors.

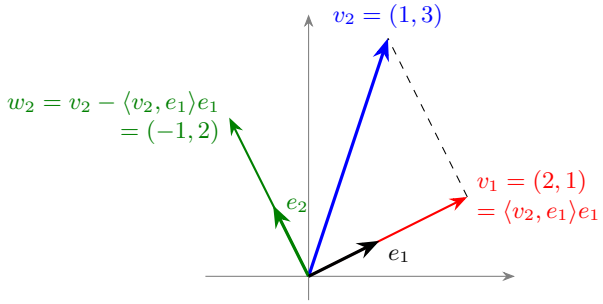
Proof. Induction on j: we show that $e_1, \dots, e_j$ is orthonormal and spans $\operatorname{span}(v_1, \dots, v_j)$.

1. $j = 1$: $v_1 \neq 0$ (it is in a basis), $\|e_1\| = \|v_1\|/\|v_1\| = 1$, and e_1 is a nonzero multiple of v_1, so the spans agree.
2. *$w_j \neq 0$, so e_j is defined.* If $w_j = 0$, then v_j is a combination of $e_1, \dots, e_{j-1}$, which span $\operatorname{span}(v_1, \dots, v_{j-1})$ by induction. That contradicts the independence of $v_1, \dots, v_j$.
3. *e_j is orthogonal to the earlier vectors.* For $k < j$, using $\langle e_i, e_k\rangle = \delta_{ik}$ for $i, k < j$:

$$\langle w_j, e_k\rangle = \langle v_j, e_k\rangle - \sum_{i<j} \langle v_j, e_i\rangle\langle e_i, e_k\rangle = \langle v_j, e_k\rangle - \langle v_j, e_k\rangle = 0.$$

   Dividing by $\|w_j\|$ keeps this and makes $\|e_j\| = 1$.
4. *Spans.* e_j is a combination of v_j and $e_1, \dots, e_{j-1}$, so $\operatorname{span}(e_1, \dots, e_j) \subseteq \operatorname{span}(v_1, \dots, v_j)$. Both lists are independent with j vectors (the e's are orthonormal by step 3 and the induction hypothesis, hence independent by Prop. 85), so both spans have dimension j, and the inclusion is an equality.
5. For $j = n$ the e's span V, so they are an orthonormal basis.

□



Gram–Schmidt for $v_1 = (2,1)$, $v_2 = (1,3)$. Here $e_1 = v_1/\sqrt{5}$ and $\langle v_2, e_1\rangle = \sqrt{5}$, so the component of v_2 along e_1 is $\sqrt{5}\,e_1 = (2,1)$. The remainder $w_2 = (-1,2)$ is perpendicular to v_1 (the dashed segment from $(2,1)$ to $(1,3)$ is w_2 moved to start at $(2,1)$), and $e_2 = w_2/\sqrt{5}$.

Example 70: orthonormalizing $1, x, x^2$ on $[0,1]$. In $\mathcal{P}_2$ with $\langle p, q\rangle = \int_0^1 p(x)\overline{q(x)}\,dx$:

1. $\|1\|^2 = \int_0^1 1\,dx = 1$, so $e_1 = 1$.
2. $\langle x, 1\rangle = \int_0^1 x\,dx = \frac{1}{2}$, so $w_2 = x - \frac{1}{2}$. Its norm: substituting $t = x - \frac{1}{2}$,
$$\|w_2\|^2 = \int_0^1 \Big(x - \frac{1}{2}\Big)^2 dx = \int_{-1/2}^{1/2} t^2\,dt = \frac{1}{12}, \qquad e_2 = \sqrt{12}\Big(x - \frac{1}{2}\Big) = \sqrt{3}(2x-1).$$
3. $\langle x^2, 1\rangle = \frac{1}{3}$ and
$$\langle x^2, e_2\rangle = \sqrt{3}\int_0^1 (2x^3 - x^2)\,dx = \sqrt{3}\Big(\frac{1}{2} - \frac{1}{3}\Big) = \frac{\sqrt{3}}{6},$$
so
$$w_3 = x^2 - \frac{1}{3} - \frac{\sqrt{3}}{6}\cdot\sqrt{3}(2x-1) = x^2 - \frac{1}{3} - \frac{1}{2}(2x-1) = x^2 - x + \frac{1}{6}.$$
4. Its norm and the normalized vector:
$$\|w_3\|^2 = \int_0^1 \Big(x^2 - x + \frac{1}{6}\Big)^2 dx = \frac{1}{180}, \qquad e_3 = 6\sqrt{5}\Big(x^2 - x + \frac{1}{6}\Big) = \sqrt{5}(6x^2 - 6x + 1).$$

These are the shifted Legendre polynomials; on $[-1,1]$ the same procedure gives the Legendre polynomials (HW10 P1). Other inner products (for example with a Gaussian weight) give other classical families, such as the Hermite polynomials. They are also eigenfunctions of certain differential operators.

Consequences.

- Every finite-dimensional inner product space has an orthonormal basis: apply Gram–Schmidt to any basis.
- Every orthonormal list $e_1, \dots, e_m$ extends to an orthonormal basis. Extend it to a basis $e_1, \dots, e_m, v_{m+1}, \dots, v_n$ and apply Gram–Schmidt; the first m vectors do not change: $\langle e_j, e_k\rangle = 0$ for $k < j$, so every subtracted term is 0 and $w_j = e_j$, and $\|e_j\| = 1$.
- *Infinite dimension.* An infinite-dimensional inner product space need not have an orthonormal basis in the sense of this course (every vector a *finite* combination). If $\{e_i\}$ is such a basis, each v is a finite combination of some $e_{i_1}, \dots, e_{i_k}$, so $\langle v, e_i\rangle = 0$ for all other i: every vector has nonzero inner product with only finitely many basis vectors. For ℓ^2 (square-summable sequences, $\langle a, b\rangle = \sum a_i\bar{b}_i$) one can show that this fails for every orthonormal family; the proof needs convergence of infinite sums in ℓ^2 and is cited, not proved.

The question answered. Yes: apply Gram–Schmidt to any basis (Prop. 87).

Shortcuts

- **Gram–Schmidt without square roots**: $w_1 = v_1$ and $w_j = v_j - \sum_{k<j} \frac{\langle v_j, w_k\rangle}{\|w_k\|^2} w_k$; normalize only at the end. Reason: $\frac{\langle v_j, w_k\rangle}{\|w_k\|^2} w_k = \langle v_j, e_k\rangle e_k$ for $e_k = w_k/\|w_k\|$. Instance: Ex. 70 gives $w = 1,\ x - \frac{1}{2},\ x^2 - x + \frac{1}{6}$ with no roots until the end.
- **Symmetric intervals**: on $[-a, a]$, $\langle x^j, x^k\rangle = 0$ when $j + k$ is odd. Reason: the integrand is odd. Instance: on $[-1, 1]$, $w_2 = x$ at once (HW10 P1).
- **Check the answer**: every $\langle e_i, e_j\rangle$ must be δ_{ij}. Instance: in Ex. 70, $\langle e_2, e_1\rangle$ is the integral of $\sqrt{3}(2x - 1)$ over $[0, 1]$, which is 0.

Traps

- Subtract components along the new e's, not the old v's. For $v_1 = (1, 0, 0)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 1, 1)$ the right answer is $e_3 = (0, 0, 1)$; subtracting along v_2 instead gives

$$v_3 - \langle v_3, e_1\rangle e_1 - \langle v_3, v_2\rangle v_2 = (-2, -2, 1),$$

  which is not even orthogonal to e_1.
- Forgetting to normalize: $\langle v, x\rangle x$ is the component of v along x only when $\|x\| = 1$; for $x = (2, 0)$, $v = (1, 0)$ it gives $(4, 0)$.
- The order matters: Gram–Schmidt on $(1, 1), (1, 0)$ gives $(1, 1)/\sqrt{2}$, $(1, -1)/\sqrt{2}$; on $(1, 0), (1, 1)$ it gives e_1, e_2.

§2.3 Representing functionals by vectors (Ch. 18)

The question. Each vector v gives a linear functional $u \mapsto \langle u, v\rangle$. Are there any other linear functionals on V?

Theorem (Prop. 88 / LADR 6.42, 6.58: Riesz representation)

Let V be finite-dimensional. For every $\psi \in V'$ there is exactly one $v \in V$ with $\psi(u) = \langle u, v\rangle$ for all u. If $e_1, \ldots, e_n$ is an orthonormal basis, this vector is

$$v = \overline{\psi(e_1)}\, e_1 + \cdots + \overline{\psi(e_n)}\, e_n$$

(over $\mathbb{R}$, no conjugates). Equivalently, the map $\Gamma\colon V \to V'$, $\Gamma(v)(u) = \langle u, v\rangle$, is a bijection. It is real-linear; over $\mathbb{C}$ it satisfies $\Gamma(\lambda v) = \bar{\lambda}\Gamma(v)$.

The idea

Γ is injective, because $\Gamma(v)$ applied to v itself is $\|v\|^2$, which is nonzero for $v \neq 0$. An injective map between spaces of the same dimension is onto. For the formula: if $v = \sum a_j e_j$, then $\psi(e_k) = \langle e_k, v\rangle = \bar{a}_k$ by conjugate-linearity in the second slot, so $a_k = \overline{\psi(e_k)}$.

Proof. 1. $\Gamma(v)$ *is a linear functional:* $u \mapsto \langle u, v\rangle$ is linear in u by (T1).

2. Γ *is additive and* $\Gamma(\lambda v) = \bar{\lambda}\Gamma(v)$, by conjugate-linearity in the second slot. In particular Γ is linear over $\mathbb{R}$, where $\bar{\lambda} = \lambda$.

3. *Injective.* If $\Gamma(v) = 0$, then $\|v\|^2 = \langle v, v\rangle = \Gamma(v)(v) = 0$, so $v = 0$.

4. *Onto.* Over $\mathbb{R}$, $\dim V' = \dim V$, so the injective linear map Γ is onto (T5). Over $\mathbb{C}$, regard V and V' as real vector spaces: if $v_1, \ldots, v_n$ is a complex basis, then $v_1, iv_1, \ldots, v_n, iv_n$ is a real basis (a real relation $\sum(a_j v_j + b_j iv_j) = 0$ is the complex relation $\sum(a_j + ib_j)v_j = 0$), so the real dimension is $2n$, and the same holds for V' (T3). So both have real dimension $2n$, and Γ is real-linear and injective, so it is onto.

5. *Formula.* The vector $v = \sum_j \overline{\psi(e_j)} e_j$ satisfies, for each k,

$$\langle e_k, v\rangle = \sum_j \psi(e_j)\langle e_k, e_j\rangle = \psi(e_k)$$

(the conjugate of $\overline{\psi(e_j)}$ comes out of the second slot as $\psi(e_j)$). So the linear functionals $u \mapsto \langle u, v\rangle$ and ψ agree on a basis, hence everywhere.

□

Example: representing $p \mapsto p(1/2)$. On $\mathcal{P}_1(\mathbb{R})$ with the L^2 product on $[0,1]$, take the functional $\psi(p) = p(1/2)$. With the orthonormal basis $e_1 = 1$, $e_2 = \sqrt{3}(2x-1)$ of Ex. 70, $\psi(e_1) = 1$ and $\psi(e_2) = 0$, so $v = 1$. Check: for every linear p,

$$\langle p, 1\rangle = \int_0^1 p(x)\,dx = p\big(\tfrac{1}{2}\big),$$

because the midpoint rule is exact for lines.

The question answered. No: in finite dimension every linear functional is $u \mapsto \langle u, v\rangle$ for exactly one v (Prop. 88).

Shortcuts

- **Riesz vector from an orthonormal basis**: $v = \sum \overline{\psi(e_j)} e_j$. Reason: Prop. 88. Instance: $\psi(p) = p(\frac{1}{2})$ on $\mathcal{P}_1$ gives $v = 1$.
- **Riesz vector by unknowns**: write $v = \sum c_j u_j$ in any basis and impose $\langle u_k, v\rangle = \psi(u_k)$ for each k; this is a linear system. Reason: two functionals that agree on a basis are equal. Instance: $\psi(p) = p(\frac{1}{2})$ on $\mathcal{P}_1$, $v = a + bx$:

$$\langle 1, v\rangle = a + \frac{b}{2} = 1, \qquad \langle x, v\rangle = \frac{a}{2} + \frac{b}{3} = \frac{1}{2},$$

so $a = 1$, $b = 0$.

Traps

- Over $\mathbb{C}$ the coefficients are conjugated (the notes omit the conjugate). On $\mathbb{C}$ with $\langle z, w\rangle = z\bar{w}$, $\psi(z) = iz$ is represented by $v = -i$, not by $\psi(1) = i$: $\langle z, i\rangle = -iz$.
- Γ is not complex-linear: $\Gamma(iv) = -i\Gamma(v)$.

§2.4 Orthogonal complements (Ch. 18)

The question. In $\mathbb{R}^3$ the vectors orthogonal to $(1,1,1)$ form the plane $x + y + z = 0$, and $(1,2,3) = (2,2,2) + (-1,0,1)$ splits into a multiple of $(1,1,1)$ and a vector in that plane. Does every subspace U give such a splitting of every vector, and what is the set of vectors orthogonal to that plane?

Definition (Def. 47 / LADR 6.47)**.** For a subspace U of V, the **orthogonal complement** is

$$U^\perp = \{v \in V : \langle v, u\rangle = 0 \text{ for all } u \in U\}.$$

- $U^\perp$ is a subspace: if $v, w \in U^\perp$ and $u \in U$, then $\langle \lambda v + w, u\rangle = \lambda \cdot 0 + 0 = 0$.
- In terms of Week 5: $v \in U^\perp$ exactly when the functional $\langle \cdot, v\rangle$ vanishes on U, that is, when $\Gamma(v) \in U^0$. So Γ carries $U^\perp$ into U^0, and onto it: every $\varphi \in U^0$ is $\Gamma(v)$ for some v (Prop. 88), and then $v \in U^\perp$ by the previous sentence.
- $\{0\}^\perp = V$ (every v has $\langle v, 0\rangle = 0$) and $V^\perp = \{0\}$ (a $v \in V^\perp$ is orthogonal to itself, so $v = 0$).
- If $U \subseteq W$, then $W^\perp \subseteq U^\perp$: a vector orthogonal to every vector of W is in particular orthogonal to every vector of U.

- *Computing it in* $\mathbb{R}^n$. If $U = \operatorname{span}(u_1, \dots, u_k)$, then $v \in U^\perp$ if and only if $\langle v, u_i \rangle = 0$ for each i (then v is orthogonal to every combination). So $U^\perp$ is the null space of the matrix whose rows are $u_1, \dots, u_k$.

Example: a complement in $\mathbb{R}^3$. For $U = \operatorname{span}\big((1,1,1)\big)$ in $\mathbb{R}^3$, $U^\perp$ is the plane $x + y + z = 0$, with basis $(-1,1,0), (-1,0,1)$.

Theorem (Prop. 89 / LADR 6.49: V splits into U and $U^\perp$)
If U is a finite-dimensional subspace of V, then $V = U \oplus U^\perp$. If V is finite-dimensional, $\dim U^\perp = \dim V - \dim U$.

The idea

Two things are needed. Nothing nonzero is in both, because such a vector would be orthogonal to itself. And every v is a sum: take an orthonormal basis $e_1, \dots, e_m$ of U and subtract from v its components along the e_j, as in Gram–Schmidt. The part removed lies in U, and what remains is orthogonal to U.

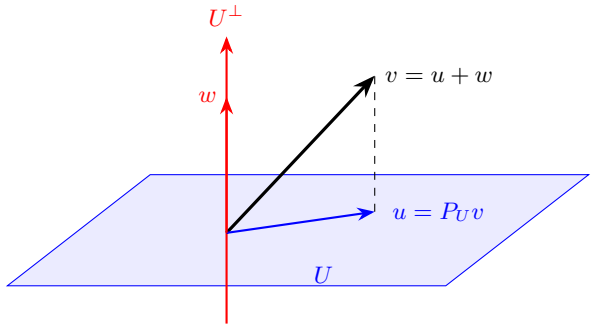
Proof. 1. $U \cap U^\perp = \{0\}$. If u is in both, then u is orthogonal to itself: $\|u\|^2 = \langle u, u \rangle = 0$, so $u = 0$.

2. $U + U^\perp = V$. U is finite-dimensional, so it has an orthonormal basis $e_1, \dots, e_m$ (Gram–Schmidt). For $v \in V$ let $u = \sum_j \langle v, e_j \rangle e_j \in U$ and $w = v - u$.

3. For each k,
$$\langle w, e_k \rangle = \langle v, e_k \rangle - \sum_j \langle v, e_j \rangle \langle e_j, e_k \rangle = \langle v, e_k \rangle - \langle v, e_k \rangle = 0.$$
So w is orthogonal to every e_k, hence to every vector of U (for $u' = \sum c_k e_k$, $\langle w, u' \rangle = \sum \bar{c}_k \langle w, e_k \rangle = 0$): $w \in U^\perp$, and $v = u + w$.

4. *Dimension.* By steps 1–3 and (T4), $\dim V = \dim U + \dim U^\perp$.

□



A plane U in $\mathbb{R}^3$ and its orthogonal complement, the perpendicular line. Each v splits as $u + w$: u is the foot of the perpendicular from v to the plane, and w points along the line. Here $\dim U + \dim U^\perp = 2 + 1 = 3$.

Consequently, for finite-dimensional V, $U^\perp = \{0\}$ if and only if $U = V$: if $U^\perp = \{0\}$, then $V = U \oplus \{0\} = U$; conversely $V^\perp = \{0\}$.

Theorem (Prop. 90 / LADR 6.52: the complement of the complement)
If U is a finite-dimensional subspace of V, then $(U^\perp)^\perp = U$.

The idea

$U \subseteq (U^\perp)^\perp$ is the definition. For the reverse, split $v \in (U^\perp)^\perp$ as $u + w$ (Prop. 89). The piece w lies in $U^\perp$, but it is also orthogonal to $U^\perp$, because v and u are. A vector orthogonal to itself is 0.

Proof. 1. $U \subseteq (U^\perp)^\perp$: every $u \in U$ is orthogonal to every vector of $U^\perp$, by the definition of $U^\perp$.

2. Let $v \in (U^\perp)^\perp$. By Prop. 89, $v = u + w$ with $u \in U$, $w \in U^\perp$. Then $w = v - u$ is in $(U^\perp)^\perp$ too, because both v and u are (step 1).

3. So w lies in $U^\perp$ and is orthogonal to every vector of $U^\perp$, including itself. Hence $\|w\|^2 = 0$, $w = 0$, and $v = u \in U$. □

Two more rules for subspaces of a finite-dimensional V:

- $(U + W)^\perp = U^\perp \cap W^\perp$. If v is orthogonal to every $u + w$, then taking $w = 0$ shows $v \perp u$ for all $u \in U$, and taking $u = 0$ shows $v \perp w$ for all $w \in W$. Conversely, if $v \perp U$ and $v \perp W$, then $\langle v, u + w\rangle = \langle v, u\rangle + \langle v, w\rangle = 0$.
- $(U \cap W)^\perp = U^\perp + W^\perp$ (HW10 P4). Apply $(U + W)^\perp = U^\perp \cap W^\perp$ to $U^\perp, W^\perp$: $(U^\perp + W^\perp)^\perp = (U^\perp)^\perp \cap (W^\perp)^\perp = U \cap W$ (Prop. 90). Taking $^\perp$ of both sides and using Prop. 90 for the subspace $U^\perp + W^\perp$ gives $U^\perp + W^\perp = (U \cap W)^\perp$.

The question answered. Yes: $V = U \oplus U^\perp$ for every finite-dimensional U (Prop. 89), and the vectors orthogonal to the plane $U^\perp$ form $(U^\perp)^\perp = U = \operatorname{span}((1, 1, 1))$ (Prop. 90).

Shortcuts

- $U^\perp$ **in** $\mathbb{F}^n$: the null space of the matrix whose rows are the conjugated spanning vectors of U; check $\dim U^\perp = n - \dim U$. Reason: $\langle v, u_i\rangle = \sum_j v_j \overline{u_{ij}}$, and Prop. 89. Instance: $U = \operatorname{span}((1, 1, 1))$ gives the plane $x + y + z = 0$.
- $U^\perp$ **in a function space**: unknown coefficients, one equation $\langle v, u_i\rangle = 0$ per spanning vector. Reason: the bullet "Computing it" above. Instance: $U = \operatorname{span}(1)$ in $\mathcal{P}_1$ with $\int_0^1$:

$$\langle a + bx, 1\rangle = \int_0^1 (a + bx)\, dx = a + \frac{b}{2} = 0$$

  gives $U^\perp = \operatorname{span}(2x - 1)$.
- **Rules**: $(U + W)^\perp = U^\perp \cap W^\perp$, $(U \cap W)^\perp = U^\perp + W^\perp$, $U \subseteq W \Rightarrow W^\perp \subseteq U^\perp$, $(U^\perp)^\perp = U$. Reason: the bullets above and Prop. 90. Instance: in $\mathbb{R}^3$, $(\operatorname{span}(e_1) + \operatorname{span}(e_2))^\perp = \operatorname{span}(e_3)$.

Traps

- $U^\perp$ depends on the inner product: for $\langle x, y\rangle = x_1y_1 + x_1y_2 + x_2y_1 + 2x_2y_2$ on $\mathbb{R}^2$ and $U = \operatorname{span}(e_1)$, $\langle e_1, y\rangle = y_1 + y_2$, so $U^\perp = \operatorname{span}((1, -1))$, not $\operatorname{span}(e_2)$.
- A complement is not the orthogonal complement: $\operatorname{span}((1, 1)) \oplus \operatorname{span}(e_1) = \mathbb{R}^2$, but $(1, 1) \not\perp e_1$.
- In infinite dimension $V = U \oplus U^\perp$ and $(U^\perp)^\perp = U$ can fail: in ℓ^2 let U be the sequences with finitely many nonzero entries. U contains every standard vector ε_k, and $\langle v, \varepsilon_k\rangle = v_k$, so $U^\perp = \{0\}$ and $(U^\perp)^\perp = \ell^2 \neq U$ (for example $(1, \frac{1}{2}, \frac{1}{4}, \dots) \notin U$).

§2.5 Projections (Ch. 18)

The question. $P(x, y) = (0, y)$ and $Q(x, y) = (x + y, 0)$ both satisfy $P^2 = P$. What does $P^2 = P$ say about how V splits, and which such maps send each vector straight to the nearest point of the range?

Theorem (Prop. 91: a projector splits V)
If $P \in \mathcal{L}(V)$ satisfies $P^2 = P$, then $V = \operatorname{null} P \oplus \operatorname{range} P$, and P fixes every vector of $\operatorname{range} P$.

The idea

Write $v = Pv + (v - Pv)$. The first piece is in $\operatorname{range} P$, and applying P to the second gives $Pv - P^2v = 0$, so it is in $\operatorname{null} P$. A vector in both is fixed by P (it is in the range) and sent to 0 (it is in the null space), so it is 0.

Proof. 1. *P fixes its range:* if $w = Pu$, then $Pw = P^2u = Pu = w$.

2. *Sum.* For $v \in V$, $v = (v - Pv) + Pv$, where $P(v - Pv) = Pv - P^2v = 0$ and $Pv \in \operatorname{range} P$.

3. *Direct.* If $w \in \operatorname{null} P \cap \operatorname{range} P$, then $w = Pw$ by step 1 and $Pw = 0$, so $w = 0$.

□

Example: a projector and a nilpotent. $P(x, y) = (0, y)$ on $\mathbb{R}^2$ has $P^2 = P$, $\operatorname{null} P$ the x-axis and $\operatorname{range} P$ the y-axis. By contrast, the nilpotent $N(x, y) = (y, 0)$ has $N^2 = 0$, $\operatorname{null} N = \operatorname{range} N$ (the x-axis), and no such splitting.

Definition (Def. 48 / LADR 6.55: orthogonal projection)**.** If U is a finite-dimensional subspace, $V = U \oplus U^\perp$ (Prop. 89), and the **orthogonal projection** onto U is $P_U(u + w) = u$ for $u \in U$, $w \in U^\perp$. It is also called the projection onto U along $U^\perp$.

Proposition (LADR 6.57: properties of P_U)
P_U is linear, $P_U^2 = P_U$, $\operatorname{range} P_U = U$, $\operatorname{null} P_U = U^\perp$, $v - P_U v \in U^\perp$ and $\|P_U v\| \le \|v\|$ for every v. It is the only projector with range U and null space $U^\perp$.

Proof. 1. *P_U is linear.* If $v = u + w$ and $v' = u' + w'$ ($u, u' \in U$, $w, w' \in U^\perp$), then $\lambda v + v' = (\lambda u + u') + (\lambda w + w')$ with $\lambda u + u' \in U$ and $\lambda w + w' \in U^\perp$; by uniqueness of the decomposition (T4), $P_U(\lambda v + v') = \lambda u + u'$.

2. *$P_U u = u$ for $u \in U$ and $P_U w = 0$ for $w \in U^\perp$:* write $u = u + 0$ and $w = 0 + w$. Hence $\operatorname{range} P_U = U$, $\operatorname{null} P_U = U^\perp$, and $P_U^2 = P_U$, since $P_U v \in U$ is fixed.

3. $v - P_U v = w \in U^\perp$, and by Pythagoras $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2$, so $\|P_U v\| \le \|v\|$.

4. *Uniqueness.* If $P^2 = P$, $\operatorname{range} P = U$ and $\operatorname{null} P = U^\perp$, then for $v = u + w$, $Pv = Pu + Pw = u + 0$, because P fixes its range (Prop. 91). So $P = P_U$.

□

The map $P(x, y) = (0, y)$ of the example is the orthogonal projection onto $\operatorname{span}(e_2)$: $(x, y) = (0, y) + (x, 0)$ with $(x, 0) \perp e_2$.

Theorem (Prop. 92 / LADR 6.57(i): the projection formula)
If $e_1, \dots, e_m$ is an orthonormal basis of U, then $P_U v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_m \rangle e_m$.

Proof. 1. Write $v = u + w$ with $u \in U$, $w \in U^\perp$; then $P_U v = u$.

2. For each k, $\langle w, e_k \rangle = 0$ because $e_k \in U$ and $w \in U^\perp$, so

$$\langle v, e_k \rangle = \langle u, e_k \rangle + \langle w, e_k \rangle = \langle u, e_k \rangle.$$

3. By Prop. 86 applied in U, and step 2,

$$u = \sum_k \langle u, e_k \rangle e_k = \sum_k \langle v, e_k \rangle e_k.$$

(This is the vector u constructed in the proof of Prop. 89.) □

For a single nonzero vector x, the orthonormal basis $e = x/\|x\|$ of $\operatorname{span}(x)$ gives

$$P_{\operatorname{span}(x)} v = \langle v, e \rangle \, e = \frac{\langle v, x \rangle}{\|x\|^2} \, x.$$

Example: projecting onto a line. Projecting $v = (1, 2, 3)$ onto $\operatorname{span}\big((1, 1, 1)\big)$ gives $\frac{6}{3}(1, 1, 1) = (2, 2, 2)$, and the remainder $(-1, 0, 1)$ is orthogonal to $(1, 1, 1)$.

Example 71. Project x^3 onto $U = \operatorname{span}(1, x)$ in $\mathcal{P}_3(\mathbb{R})$, with the L^2 product on $[0, 1]$. The orthonormal basis $1, \sqrt{3}(2x-1)$ of U comes from Ex. 70. Then $\langle x^3, 1\rangle = \frac{1}{4}$ and

$$\langle x^3, \sqrt{3}(2x-1)\rangle = \sqrt{3}\Big(\frac{2}{5} - \frac{1}{4}\Big) = \frac{3\sqrt{3}}{20},$$

so

$$P_U(x^3) = \frac{1}{4}\cdot 1 + \frac{3\sqrt{3}}{20}\cdot\sqrt{3}(2x-1) = \frac{1}{4} + \frac{9}{20}(2x-1) = \frac{9}{10}x - \frac{1}{5}.$$

Check: the remainder $x^3 - \frac{9}{10}x + \frac{1}{5}$ has integral 0 against 1 and against x on $[0, 1]$.

The question answered. $P^2 = P$ says $V = \operatorname{null} P \oplus \operatorname{range} P$ with P fixing the range (Prop. 91). The orthogonal projections, those with $\operatorname{null} P \perp \operatorname{range} P$, send each vector to the nearest point of the range (next section): P is one, Q is not.

Shortcuts

- **Projection onto a line**: $P_{\operatorname{span}(x)}v = \frac{\langle v,x\rangle}{\|x\|^2}x$; no need to normalize. Reason: Prop. 92 with $e = x/\|x\|$. Instance: $(1,2,3) \mapsto (2,2,2)$ for $x = (1,1,1)$.
- **Bigger subspace**: orthonormalize a basis of U (Gram–Schmidt), then add the line projections (Prop. 92). Instance: Ex. 71.
- **Small complement**: $P_{U^\perp} = I - P_U$; so to project onto a plane in $\mathbb{R}^3$ with normal n, use $v - \frac{\langle v,n\rangle}{\|n\|^2}n$. Reason: $v = P_U v + (v - P_U v)$ is the splitting of v along $U^\perp \oplus (U^\perp)^\perp$ (Prop. 90). Instance: onto $x+y+z=0$: $(1,2,3) \mapsto (1,2,3) - (2,2,2) = (-1,0,1)$.
- **Matrix of P_U in $\mathbb{F}^n$**: QQ^*, where the columns of Q are an orthonormal basis of U (Q^* is the conjugate transpose). Reason: $QQ^*v = \sum_j \langle v, e_j\rangle e_j$. Instance: $U = \operatorname{span}((1,1,1))$ gives the matrix with every entry $\frac{1}{3}$.
- **Check**: $v - P_U v$ must be orthogonal to each spanning vector of U. Reason: Prop. 89. Instance: the check in Ex. 71.

Traps

- Line projections onto a non-orthogonal basis do not add up to P_U: in $\mathbb{R}^2$ with $U = \mathbb{R}^2$, basis $(1,0), (1,1)$, $v = (0,1)$, the line projections are $(0,0)$ and $(\frac{1}{2}, \frac{1}{2})$, whose sum is not $P_U v = v$.
- $P^2 = P$ does not make P an orthogonal projection: $Q(x,y) = (x+y, 0)$ has $\|Q(1,1)\| = 2 > \sqrt{2}$.

§2.6 Best approximation (LADR 6.61, HW10 P5)

The question. Which line $a + bx$ is closest to x^3 on $[0, 1]$, in the sense of making $\int_0^1 (x^3 - a - bx)^2\,dx$ as small as possible?

Theorem (Orthogonal projection is the closest point)

If U is a finite-dimensional subspace and $v \in V$, then $\|v - P_U v\| \le \|v - u\|$ for every $u \in U$, with equality only for $u = P_U v$.

The idea

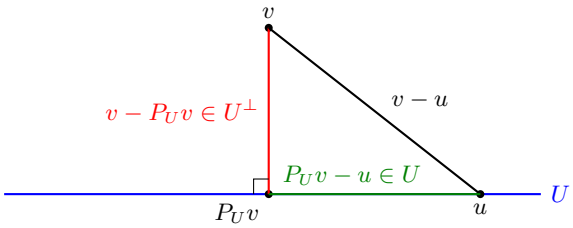
Go from v to a point u of U in two legs: first from v to $P_U v$, perpendicular to U, then from $P_U v$ to u, inside U. The two legs are perpendicular, so by Pythagoras the direct distance is longer than the first leg unless the second leg is 0.

Proof. 1. $v - u = (v - P_U v) + (P_U v - u)$. The first vector is in $U^\perp$ and the second is in U (both $P_U v$ and u are), so they are orthogonal.

2. By Pythagoras,
$$\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \ge \|v - P_U v\|^2.$$

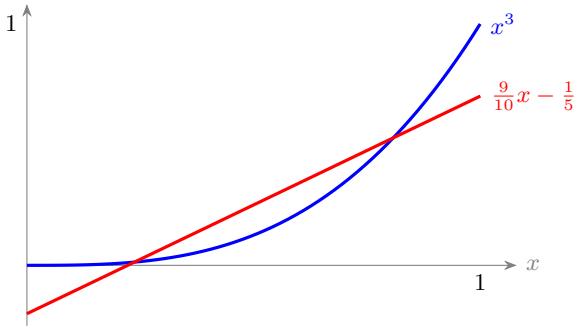
3. Equality holds exactly when $\|P_U v - u\| = 0$, that is, $u = P_U v$.

□



For any candidate $u \in U$, the three points form a right triangle, so $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2$. Moving u away from $P_U v$ only adds to the distance.

Example 71 as an approximation. Among all lines $a + bx$, the line $\frac{9}{10}x - \frac{1}{5}$ makes $\int_0^1 \big(x^3 - a - bx\big)^2 dx$ smallest. The minimum is $\|x^3 - P_U(x^3)\|^2 = \frac{9}{700}$, so the root-mean-square error (the square root of this minimum; on an interval of length 1 it is the root of the average squared error) is $\frac{3\sqrt{7}}{70} \approx 0.113$.



x^3 and its best approximation by a line in the L^2 sense on $[0, 1]$. The line is below the curve near the ends and above it in the middle, so that the error is orthogonal to 1 (it averages to 0) and to x.

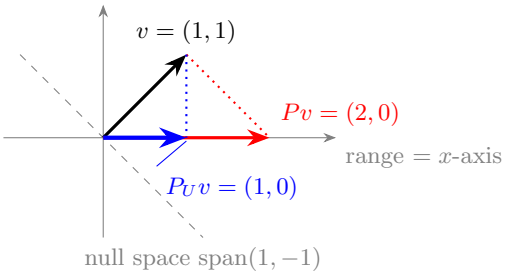
Least squares. The same theorem fits data. To find the line $y = a + bt$ closest to the points (t_i, y_i) in the sense of minimizing $\sum(y_i - a - bt_i)^2$, project $y = (y_1, \dots, y_k)$ onto $U = \operatorname{span}\big((1, \dots, 1), (t_1, \dots, t_k)\big)$ in $\mathbb{R}^k$. The projection is characterized by the remainder being orthogonal to both spanning vectors, namely $\langle y - a\mathbf{1} - bt, \mathbf{1}\rangle = 0$ and $\langle y - a\mathbf{1} - bt, t\rangle = 0$, two linear equations for a, b (Shortcuts below).

Example: an oblique projector. A projector need not be orthogonal. $P(x, y) = (x + y, 0)$ on $\mathbb{R}^2$ satisfies $P^2 = P$, with range the x-axis and null space $\operatorname{span}\big((1, -1)\big)$, which is not perpendicular to the range. It can lengthen vectors:
$$\|P(1, 1)\| = \|(2, 0)\| = 2 > \sqrt{2} = \|(1, 1)\|.$$

An orthogonal projection never does, because

$$\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2 \geq \|P_U v\|^2;$$

HW10 P7 shows this property characterizes orthogonal projections.



The oblique projector P moves v parallel to the null space span$(1,-1)$ (red dotted line) until it reaches the x-axis at $(2,0)$, which is longer than v. The orthogonal projection P_U onto $U =$ the x-axis moves v straight down to $(1,0)$ (blue dotted).

The question answered. $P_U(x^3) = \frac{9}{10}x - \frac{1}{5}$ (Ex. 71), by the theorem that orthogonal projection is the closest point.

Shortcuts

- **Closest point / minimize $\int(\dots)^2$**: the answer is $P_U v$; the minimum value is $\|v\|^2 - \sum_j |\langle v, e_j\rangle|^2$ for an orthonormal basis e_j of U. Reason: Pythagoras, $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2$, and Prop. 84. Instance: Ex. 71, $\frac{1}{7} - \left(\frac{1}{16} + \frac{27}{400}\right) = \frac{9}{700}$.
- **Least squares by normal equations**: for the line $a + bt$ through k points (t_i, y_i), solve

$$ak + b\sum t_i = \sum y_i, \qquad a\sum t_i + b\sum t_i^2 = \sum t_i y_i.$$

  Reason: the remainder $y - a\mathbf{1} - bt$ is orthogonal to $\mathbf{1} = (1,\dots,1)$ and to t. Instance: for the points $(0,0), (1,1), (2,1)$, $3a + 3b = 2$ and $3a + 5b = 3$, so $b = \frac{1}{2}$, $a = \frac{1}{6}$.

Traps

- The formula $\|v\|^2 - \sum |\langle v, u_j\rangle|^2$ needs an orthonormal basis: in $\mathbb{R}^2$ with $U = \mathbb{R}^2$, basis $(2,0), (0,1)$, $v = (1,0)$, it gives $1 - 4 = -3$, while the distance is 0.
- The best L^2 line is not the tangent line: the tangent to x^3 at $\frac{1}{2}$ is $\frac{3}{4}x - \frac{1}{4}$, the best line is $\frac{9}{10}x - \frac{1}{5}$.

§3 Problem set

HW10 P1. On $\mathcal{P}_2(\mathbb{R})$ use the inner product $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$. Apply the Gram–Schmidt procedure to the ordered basis $1, x, x^2$ to obtain an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

Hint: Prop. 87; on $[-1,1]$, $\langle x^j, x^k\rangle = 0$ when $j + k$ is odd, so most coefficients are 0.

HW10 P2. Let $W = \operatorname{span}\big((1,-1,1,0), (0,2,1,0)\big) \subseteq \mathbb{R}^4$ with the standard inner product. Find a basis of $W^\perp$.

Hint: $W^\perp$ is the null space of the 2×4 matrix whose rows span W; $\dim W^\perp = 2$ (Prop. 89).

HW10 P3. On $\mathcal{P}_2(\mathbb{R})$ use the same inner product as in Problem 1, and let $W = \operatorname{span}(x + 1)$. Find a basis of $W^\perp$.

Hint: write $p = a + bx + cx^2$ and impose $\langle p, x + 1\rangle = 0$; expect $\dim W^\perp = 2$.

HW10 P4. Let U and W be subspaces of a finite-dimensional inner-product space V. Prove $(U \cap W)^\perp = U^\perp + W^\perp$.

Hint: apply $(A + B)^\perp = A^\perp \cap B^\perp$ to $A = U^\perp$, $B = W^\perp$, then Prop. 90 twice.

HW10 P5. Let U be a subspace of a finite-dimensional inner-product space V, and let $e_1, \dots, e_m$ be an orthonormal basis of U.

(a) Prove that $P_U v = \sum_{j=1}^{m} \langle v, e_j \rangle e_j$.

(b) Prove the best-approximation property $\|v - P_U v\| \le \|v - u\|$ for every $u \in U$, and determine when equality holds.

Hint: (a) Prop. 92; (b) Pythagoras for $v - u = (v - P_U v) + (P_U v - u)$.

HW10 P6. (optional) On $\mathcal{P}_1(\mathbb{R})$ use the inner product $\langle f, g \rangle = \int_{-1}^{1} f(x) g(x)\, dx$. Find the unique polynomial $p \in \mathcal{P}_1(\mathbb{R})$ such that $\int_0^2 f(x)\, dx = \langle f, p \rangle$ for every $f \in \mathcal{P}_1(\mathbb{R})$.

Hint: Prop. 88 with the orthonormal basis $\frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}}x$ from HW10 P1, or write $p = a + bx$ and test $f = 1$ and $f = x$.

HW10 P7. (optional) Let P be a projector on a finite-dimensional inner-product space, so $P^2 = P$. Prove that P is an orthogonal projection if and only if $\|Pv\| \le \|v\|$ for every v.

Hint: ($\Rightarrow$) $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2$; ($\Leftarrow$) apply the inequality to $u + tw$ with $u \in \operatorname{range} P$, $w \in \operatorname{null} P$ and small t, to get $\langle u, w \rangle = 0$.

O1 (best quadratic error). Find the polynomial $p \in \mathcal{P}_1(\mathbb{R})$ that minimizes $\int_0^1 \big(x^2 - p(x)\big)^2 dx$, and the minimum value.

Hint: the minimizer is $P_{\mathcal{P}_1}(x^2)$; use the orthonormal basis $1, \sqrt{3}(2x - 1)$ of Ex. 70.

O2 (distance to a line). In $\mathbb{R}^4$ with the dot product let $U = \operatorname{span}\big((1,1,1,1)\big)$ and $v = (1,2,3,4)$. Find $P_U v$, a basis of $U^\perp$, and the distance from v to U.

Hint: projection onto a line; $U^\perp$ is the null space of (1 1 1 1).

O3 (reflections). Let U be a subspace of a finite-dimensional inner product space and $R = I - 2P_U$. Prove that $\|Rv\| = \|v\|$ for all v and that $R^2 = I$. Describe R geometrically when U is a line in $\mathbb{R}^2$.

Hint: write $v = u + w$ with $u \in U$, $w \in U^\perp$ and compute Rv.

O4 (equality in Bessel). Let $e_1, \dots, e_m$ be an orthonormal list in V, not necessarily a basis. Prove that equality holds in Bessel's inequality $\sum_{j=1}^{m} |\langle v, e_j \rangle|^2 \le \|v\|^2$ exactly when $v \in \operatorname{span}(e_1, \dots, e_m)$.

Hint: the proof of Bessel's inequality: equality means the remainder w is 0.

O5 (least squares). Find the line $y = a + bt$ that minimizes $\sum_i (y_i - a - bt_i)^2$ for the points $(0,1)$, $(1,2)$, $(2,4)$.

Hint: the remainder must be orthogonal to $(1,1,1)$ and $(0,1,2)$ (normal equations).

O6 (representing evaluation). On $\mathcal{P}_2(\mathbb{R})$ with $\langle p, q \rangle = \int_0^1 pq$, find the polynomial v with $\langle p, v \rangle = p(0)$ for every $p \in \mathcal{P}_2(\mathbb{R})$.

Hint: Prop. 88 with the orthonormal basis of Ex. 70: $v = \sum e_j(0) e_j$.